Rep. Proportion $\frac{150}{250} = 60\%$ (part on all); evolution 250 $\frac{-200}{200} = 25\%$ (relative change).

4. Back · Table [IMG] Table of values of f :

x	0	1	2	3
$f(x)$	4	1	0	1

Image of 2 ? Preimage of 1 ?

Rep. $f(2) = 0$; preimage of 1:1 and 3.

5. Red wire · MCQ [REP] On the globe, two points of the same parallel have:

A. the same latitude **B.** the same longitude **C.** the same distance at Greenwich **D.** the same altitude

Rep. A. the same latitude

Latitude: north/south angle to equator; longitude: east/west angle to Greenwich meridian.

Variations and extremes of a function

Form 43 Thu. 11 Feb. 2027 Growth, decline, monotonicity

1. Warming · Flash [IMG] $f(x) = -2x + 5$. Calculate $f(1)$ and $f(3)$; What do we see?

Rep. $f(1) = 3$, $f(3) = -1$: The order of the images is inverse to that of the abscisses (f decrease).

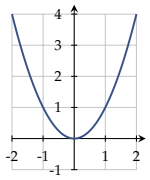
2. Heart · MCQ [MOD] f is increasing on I means that, for all $a < b$ in I :

A. $a \leqslant b$ only **B.** $f(a) \geqslant f(b)$ **C.** $f(a) \leqslant f(b)$ **D.** $f(a) = f(b)$

Rep. C. $f(a) \leqslant f(b)$

Crescent: keeps order; decreasing: reverses it.

3. Heart + · Graph [REG] Curve of $f(x) = x^2$ on $[-2;2]$:



Give variations in f.

Rep. Decreasing on $[-2;0]$, increasing on $[0;2]$.

Expression used for the route: f $(x) = x^2$.

4. Back · Flash [PRO] What multiplier coefficient corresponds to an increase of 15%? 15 % ?

Rep. 1,15 ; 0,85.

5. Red thread · Flash [IVD] Is 457 divisible by 3? By 9? Justify.

Rep. Sum of figures $= 16$, not multiple of 3: 3, or 9.

Form 44 Mon 1st March 2027 Table of variations ; maximum and minimum over an interval

1. Warming · Flash [IMG] Calculate x^2 for $x = -2, 0, 3$.

Rep. 4 ; 0 ; 9.

2. Heart · Method [REG] f is decreasing on $[-3;1]$ with $f(-3) = 5$, $f(1) = -1$, then increasing on $[1;4]$ with $f(4) = 4$. Give the maximum and minimum of f on $[-3;4]$.

Rep. Maximum 5 reached in $x = -3$ ($5 > (4)$); minimum -1 reached in $x = 1$.

Compare values at the ends of the range AND with changes in variation.

3. Heart + · True/false [REG] True or false? Justify. The maximum of a function on $[a;b]$ is still achieved in a or in b.

Rep. Wrong. Counterexample: $f(x) = -x^2 + 4$ on $[-2;2]$: maximum 4 reached in 0.

4. Back · Table [PNP] Is this table proportionate?

x	1	2	3
y	5	9	13

Rep. No: $\frac{5}{1} \neq \frac{9}{2}$ (constant increase) $+4$ but $y(0) \neq 0$).

5. Red thread · Method [GEO] Is the side triangle 5, 12 and 13 rectangle? Which property to use?

Rep. $5^2 + 12^2 = 169 = 13^2$: Converse Pythagorean theorem, rectangle (right angle opposite to 13).

Form 45 Mar. 2 March 2027 Affinities: rate of increase and changes

1. Warming · Flash [IMG] $f(x) = 3x - 1$. Calculate $\frac{f(4) - f(1)}{4 - 1}$.

Rep. $\frac{11 - 2}{3} = 3$: it's the slope m.

2. Heart · Flash [MOD] Meaning of variation of $f(x) = -4x + 7$?

Rep. $m = -4 < 0$: f is strictly decreasing on $\mathbb{R}$.

3. Heart + · Demonstration [LIT] Show that $f(x) = 5x - 2$ is increasing on $\mathbb{R}$.

Rep. For $a < b$: $f(b) - f(a) = 5(b - a) > 0$, therefore $f(a) < f(b)$.

Drafting model for the programme demonstration.

4. Back · Table [LEC] Ages of the students of a club.

Age	10	11	12	13
Count	3	6	10	6

How many students are up to 11 years old? How often?

Rep. $3 + 6 = 9$ students; $\frac{9}{25} = 36\,\%$.

5. Red thread · Programme → formula [REG] Program: multiply by 3 then subtract 2. $f(x)$ then $f(4)$.

Rep. $f(x) = 3x - 2$; $f(4) = 10$.

Form 46 4 March 2027 Absolute and square value functions: sign and variations

1. Warming · Flash [CAL] Calculate $|-5|$, $(-5)^2$ and $|2 - 7|$.

Rep. 5 ; 25 ; 5.

2. Heart · Method [REG] Give the table of changes in $x \mapsto |x|$ and his sign.

Rep. Decreasing on $]-\infty; 0]$, increasing on $[0; +\infty[$; minimum 0 in 0 ; always positive.

3. Heart + · Reflection [CMP] Compare $(-3)^2$ and $(-2)^2$ then 3^2 and 2^2. Is the square function increasing on $\mathbb{R}$?

Rep. $9 > 4$ in both cases, but $-3 < -2$: the square reverses the order on $]-\infty; 0]$; no, not increasing on $\mathbb{R}$.

4. Return · Method [PRB] We're launching two balanced dice.

Rep. 6 out of 36 favourable cases: $\frac{1}{6}$.

5. Red thread · Method [ALG] Program: choose a number; add 1; raise squared; subtract squared
What is the result for 6? What is the general expression?

Rep. For 6: $49 - 36 = 13$. General $(x + 1)^2 - x^2 = 2x + 1$.